

cuOpt benchmark: warehouse MILP

T1 · Appendix A1–A2 · GPU MILP · 316,814 variables

Internal benchmark reported by Omelet. Results below are transcribed from the exact final-pitch appendix, not newly rerun tests. They support discussion of experimental performance; they do not establish shipped NVIDIA SDK adoption. Original run logs and named test-owner confirmation are not attached; independent verification is pending.

Metric	Wall-clock time to optimality; solve time and total time reported separately.
Problem	316,814 variables and 240,120 constraints; warehouse planning/slotting.
Comparison	cuOpt 26.08 on one RTX 4090 versus SCIP 9.0, CP-SAT 9.11 and CBC 2.10 on CPU, 64 threads.
Setup	Xeon Gold 6326 x2; RTX 4090, 24 GiB (2.0 GiB used as reported); CUDA 13.0; Ubuntu 24.04. Three-run median; solutions independently re-simulated as reported.
Test date / owner	Not specified in the supplied MILP appendix. Date and responsible engineer/team: not yet verified.

SOLVER	HARDWARE	SOLVE TIME	TOTAL TIME	CPU TOTAL / CUOPT TOTAL
NVIDIA cuOpt 26.08	1 GPU	13.6 s	15.4 s	1.00×
SCIP 9.0	CPU, 64T	18.1 s	21.4 s	1.39×
CP-SAT 9.11	CPU, 64T	15.0 s	22.6 s	1.47×
CBC 2.10	CPU, 64T	40.8 s	44.1 s	2.86×

Feasibility and scaling

The appendix reports all four solvers reaching the same optimum: 1,139 pallet positions, with 0 constraint violations on re-simulation and a reported 24.3% reduction in peak space. The operational baseline behind the peak-space reduction is not specified.

For an increase from 62K to 317K variables (5.1×), reported time growth is 1.7× for cuOpt, 6.3× for CP-SAT, 7.9× for SCIP and 9.4× for CBC. At smaller, single-warehouse scale, SCIP is reported at 0.2–2.7 seconds. GPU benefit is scale-dependent.

Source and reproducibility

[Open appendix A2: full results](#) · [Open appendix A1: summary](#)

Required to reproduce: model/instance, solver commands and settings, tolerances, seeds, per-run timings, validation output and test date/owner. No fabricated run records are supplied.